

Appendix A

Project Cost Estimate

Proposed Budget for Materials

Item Description	Quantity	Cost/Unit (PHP)	Subtotal (PHP)
[1] Raspberry Pi 4/5 (Central Controller)	1	₱7,449.00	₱7,449.00
[2] ESP32 Microcontroller (Edge Feeder Node)	1	₱450.00	₱450.00
[3] 7-inch DSI Touchscreen Display	1	₱2,800.00	₱2,800.00
[4] 0.18 m3 Hopper & Trough Fabrication	1	₱5,000.00	₱5,000.00
[5] Sensors & Actuators (Load Cell, Flow, MQ-135, Solenoids)	1 set	₱2,500.00	₱2,500.00
Total			₱18,249.00

Proposed Budget for Lab Equipment/Apparatus Use

Item Description	Quantity	Cost/Unit	Subtotal
[1] Reference Weights (Phase 1 Testing)	1 set	₱2,000.00	₱2,000.00
[2] MQ-135 Calibration Environment	1	₱1,500.00	₱1,500.00
Total			₱3,500.00

Room Rental Costs

Room	No. of hours	Cost/hour	Subtotal
[1] University Engineering Lab	—	₱3,200.00	₱3,200.00
[2] Piggery Field Site (Site Assessment Phase)	—	Provided by client	—
Total			₱3,200.00

Miscellaneous Costs

Manpower/Overtime Costs

Name of Person	Position	Rate/Fee	No. of Hours
[1] Manpower (Overtime Installation)	Lead/Helper	₱1,000.0/person	—
Total			₱3,000.00

External Analysis

Description	Samples	Analysis Fee	Subtotal
-------------	---------	--------------	----------

[1] Ammonia Threshold Calibration (Known Concentrations)	1 Session	₱1,500.00	₱1,500.00
Total			₱1,500.00

Documentation Costs

Description	Quantity	Cost/Unit	Subtotal
Thesis Manuscript Printing and Binding	5	₱300.00	₱1,500.00
Total			₱1,500.00

Transportation and Accommodation Costs

Description	No. of persons	Rate/person	Subtotal
Field Site Visit (Cagayan, Round Trip)	5	₱5,060.00	₱25,300.00
Total			₱25,300.00

Incidental Costs

Description	Quantity	Cost/Unit	Subtotal
RS-485 Cables & DC Power Distribution	1 lot	₱1,000.00	₱1,000.00
Total			₱1,000.00

GRAND TOTAL			₱54,249.00
--------------------	--	--	-------------------

Workplan and Expected Outputs

[illegible]

Appendix C

Notice of Adviser Acceptance



UNIVERSITY of
SAN CARLOS
SCIENTIA • VIRIUS • DEVOTIO

NOTICE OF ACCEPTANCE

We, the undersigned, hereby agree to collaborate, as part of the requirements of the Department of Computer Engineering's **BS CPE thesis courses**, in a project/research study entitled, **Integrated Automated Feeding and Watering System with Centralized Data Monitoring for Commercial Piggeries**. This Notice of Acceptance is effective beginning the **1st Semester of AY 2024-2025**.

The **students** will prepare a project/research proposal, including an introductory chapter, a review of related literature, methodology, and a proposed timetable and budget. Upon approval, the students will conduct the study as outlined in the approved methodology, document the results, discussion, conclusions, and recommendations, and defend the study before an appointed thesis evaluation committee.

The **adviser**, whose own research project(s) may provide the foundation for the students' study, is responsible for guiding the students through the project/research study, holding regular consultation sessions at a mutually agreed frequency (e.g., weekly, bimonthly), and reviewing and critiquing the thesis manuscript before it is submitted to the thesis evaluation committee for the proposal hearing and oral examination.

The **assistant adviser** will also support the students within his/her technical capacity and expertise. In the event of the adviser's absence or any circumstance that precludes the adviser from fulfilling their duties, the assistant adviser shall assume the responsibilities of thesis advising.

To formalize this agreement, we hereby affix our signatures below.


PACE, VAUN MICHAEL C.


PARRAS, PAUL ANDREW F.


SORIANO, LORENZ ANTHONY L.


ENGR. ALVIN JOSEPH S. MACAPAGAL
Thesis Adviser


ENGR. JAN DAVE Q. CAMPAÑERA
Thesis Assistant Adviser

Witnesses:


DR. GODWIN S. MONSERATE
CPE 3107 Coordinator


DR. ANTONIETTE M. CAÑETE
Chair, Department of Computer Engineering

USC - Talamban Campus
Gov. M. Cuenco Ave., Talamban, Cebu City, Philippines 6000
Email: comdept@usc.edu.ph
www.usc.edu.ph

Appendix D

Student-Adviser Thesis Counseling Logbook



**UNIVERSITY of
SAN CARLOS**
SCIENTIA • VIRTUS • DEVSOTIO

**Department of Computer Engineering
School of Engineering**

STUDENT-ADVISER THESIS COUNSELING LOGBOOK

Thesis Proposal (CPE 3207L)
Thesis Implementation (CPE 3301L | CPE 4101L)

1st Semester AY _____
2nd Semester AY 2025-2026
Summer _____

GROUP LETTER:	V
PROJECT TITLE:	Integrated Automated Feeding and Watering System with Centralized Data Monitoring for Commercial Piggeries
NAME OF ADVISER:	Engr. Alvin Joseph Macapagal
NAME OF ASST. ADVISER:	Engr. Jan Dave Q. Campañera
NAME OF STUDENTS:	Vaun Michael C. Pace
	Paul Andrew F. Parras
	Lorenz Anthony L. Soriano

DATE	VENUE / MEDIUM	TIME	TOPICS DISCUSSED	ADVISER'S / ASST. ADVISER'S SIGNATURE
12/04/2025	DML	10:15 AM - 10:30 AM	-Presented a thesis proposal, and the adviser recommended securing a validating client and identifying a gas sensor suitable for detecting swine emissions.	
12/17/2025	DML	2:00 PM - 3:00 PM	-Discussed with the adviser about conducting a feasibility study for the clients.	

USC - Talamban Campus
 Nasipit, Talamban, Cebu City, Philippines 6000
 Email: comdept@usc.edu.ph
www.usc.edu.ph



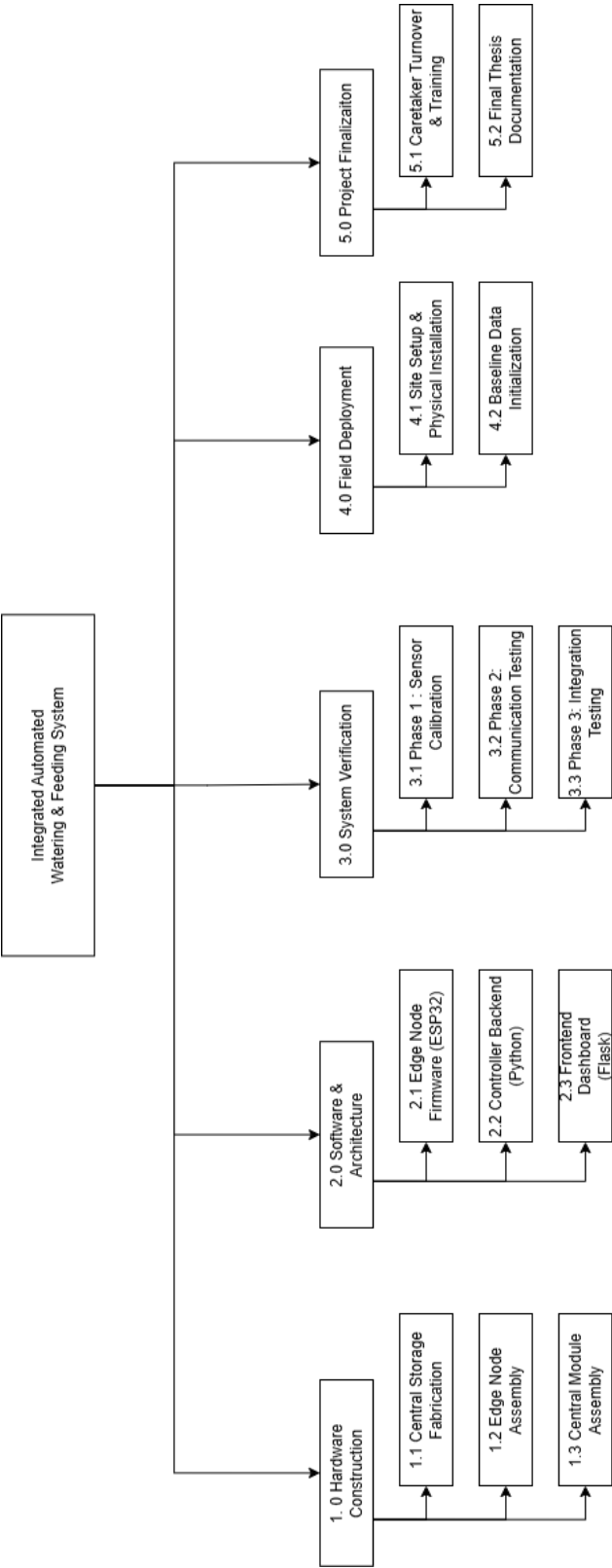
UNIVERSITY of SAN CARLOS

SCIENTIA • VIRTUS • DEVOTIO

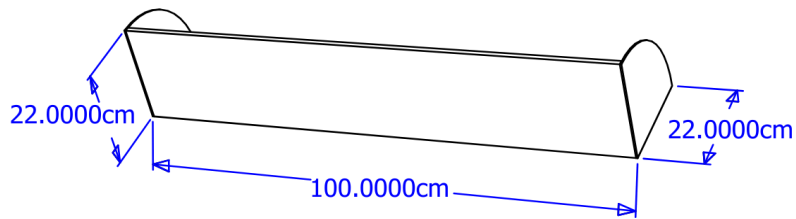
03/18/2025	DML	1:30 PM - 2:30 PM	-Begin drafting, obtain a thesis copy from Engr. Campañera. -Confirm details with the client -Review related papers to identify common elements.	
03/25/2025	DML	1:30 PM - 2:10 PM	-Clearly qualify the problems, formulate the objectives, structure the manuscript based on the required format. -Focus on completing the Introduction, Significance of the Study, and Goals and Objectives sections.	
04/08/2025	DML	1:30 PM - 2:00 PM	- Develop the methodology -Restructure the goals and objectives -Decide on testing conditions, preferably validating the system with pigs.	
04/15/2025	DML	2:00 PM - 2:15 PM	-changes to the methodology based on the adviser's recommendations	
04/27/2025	DML	1:00 PM - 1:20 PM	-Finalizing the manuscript	

USC - Talamban Campus
Nasipit, Talamban, Cebu City, Philippines 6000
Email: comedept@usc.edu.ph
www.usc.edu.ph

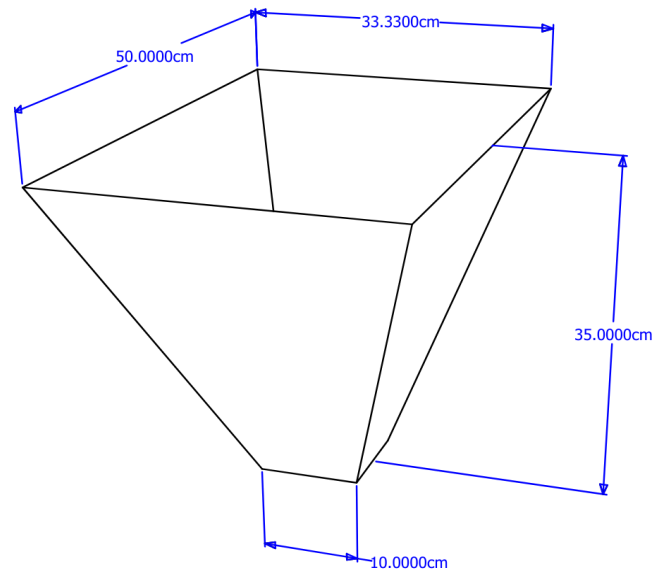
Appendix E
Work Breakdown Structure



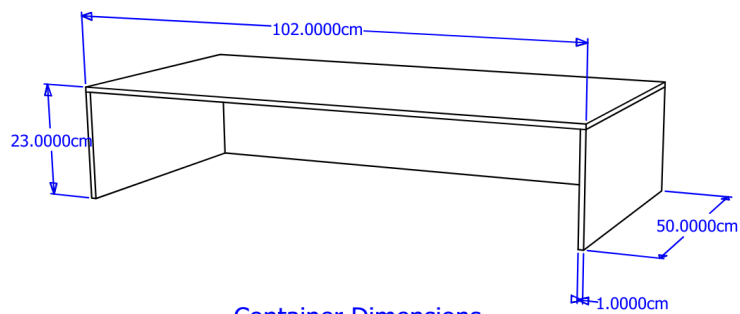
Appendix F Prototype Design



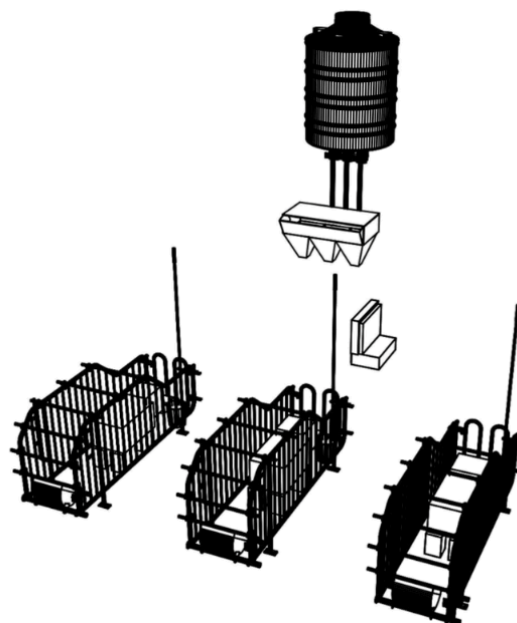
Lid Dimensions



Hopper Dimensions

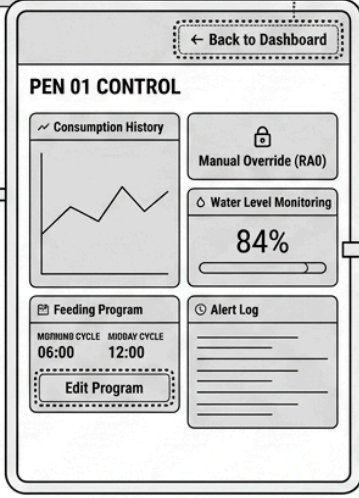
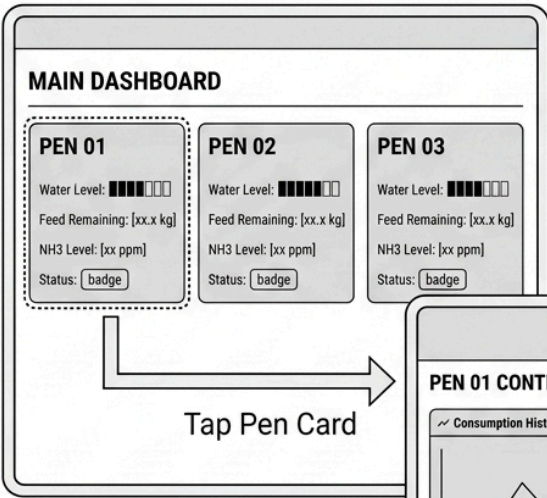


Container Dimensions



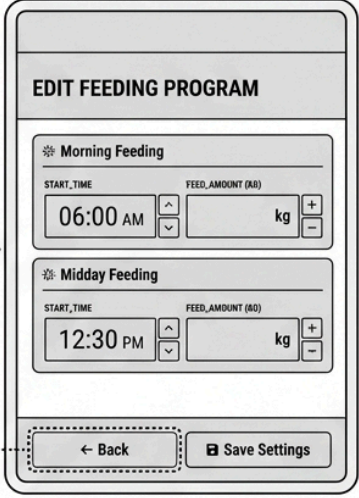
Appendix G
UI Wireframe

SCREEN 1 (LEFT)
Main Dashboard



SCREEN 2 (CENTER)
Pen Control

Tap Edit Program



SCREEN 3 (RIGHT)
Feed Settings

Back

Appendix H

Project Deliverables

#	Project Deliverables	Status
A. Hardware Deliverables		
1	Central Control Module (Raspberry Pi, Touchscreen, RS-485, LoRa, UPS)	Ongoing
2	Central Feed Hopper Assembly (0.18 m ³ Frustum Hopper with Guillotine Valve)	Ongoing
3	Auger Conveyor Mechanism (NEMA 17 Stepper Motor-Driven Feed Dispenser)	Ongoing
4	Feeder Module Assemblies ×3 (ESP32, HX711, Load Cell, MQ-135, JSN-SR04T)	Ongoing
5	Feed Trough Units ×3 (with Load Cell Mounts)	Ongoing
6	Water Trough & Solenoid Valve Assembly ×3 (PVC-Piped Supply Tank)	Ongoing
7	RS-485 Wired Bus Network (Twisted-Pair, MAX485 Transceivers)	Ongoing
8	LoRa-GSM Remote Alert Gateway	Ongoing
9	UPS Power Backup Module (18650 Lithium Cell Circuit)	Ongoing
10	Manual Override Push Button (Active-High, Pin RA0)	Ongoing
11	Standard Reference Weights (for Load Cell Calibration)	Ongoing
B. Software Deliverables		
12	ESP32 Edge Node Firmware (FreeRTOS Dual-Core, MODBUS RTU Slave)	Ongoing
13	Raspberry Pi Python Backend (PyModbus Poller, Command & Alert Services)	Ongoing
14	Anomaly Detection / Analytical Engine (Rolling-Window, N=14 Sessions)	Ongoing
15	SQLite Local Database (Timestamped Telemetry, Alerts, Schedules)	Ongoing

#	Project Deliverables	Status
16	Flask Web Application Backend (RESTful API, AJAX Polling)	Ongoing
17	Touchscreen Dashboard Frontend (Main Dashboard, Pen Control, Feed Settings)	Ongoing
18	LoRa-SMS Alert Pipeline (Message Queuing, LoRa Payload Encoding)	Ongoing
19	Python Watchdog Service (Heartbeat Logging, Uptime Computation)	Ongoing
C. Testing and Evaluation Deliverables		
20	Feed Dispensing Accuracy Report – T6.1 (FDA $\geq 95\%$, 30 Trials/Module)	Ongoing
21	Water Level Sensor Validation Report – T6.2 (± 1 cm, 10 Trials)	Ongoing
22	MQ-135 Air Quality Calibration Report – T6.3 (25 ppm NH ₃ Threshold)	Ongoing
23	RS-485 Signal Integrity Report – T7.1 (Differential Voltage $\geq \pm 1.5$ V)	Ongoing
24	MODBUS RTU Packet Loss Report – T7.2 (Packet Loss $\leq 1\%$)	Ongoing
25	System Uptime & Data Latency Report – T8.1 (U _{sys} $\geq 95\%$, Latency ≤ 2 s)	Ongoing
26	Alert Response & SMS Delivery Report – T8.2 (ARA $\geq 90\%$, SDR $\geq 95\%$)	Ongoing
27	LoRa Communication Range Report – T8.3 (RSSI ≥ -120 dBm at Max Distance)	Ongoing
28	Load Cell Calibration & Accuracy Report - T2.1 ($R^2 \geq 0.99$, $\pm 2\%$ accuracy)	Ongoing
29	End-to-End Alert Latency Report - T4.3 (Mean latency ≤ 30 seconds per alert)	Ongoing
D. Deployment Deliverables		
30	Site Assessment & Electrical Routing (Cable Routing Diagram, EMI Report)	Ongoing
31	Physical Installation (Hopper, Feeder Nodes, Solenoid Plumbing)	Ongoing